

Fortgeschrittenenpraktikum

Raman Spectroscopy

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Laboratory date: 31 September 2026
Report submitted: September 28, 2026

Abstract

The vibration Raman effect consists of the appearance of vibration inelastic (Stokes) and superelastic (Antistokes) lines in the fluorescence spectrum of gases, liquids and solids. In this experiment the vibration Raman spectra of various liquid substances are recorded and identified.

1 Theoretical background

1.1 Raman scattering

Vibrational spectroscopies are employed in the determination of the molecular structures and are used in many areas. Among the available techniques, Raman spectroscopy is essential in the study of nanostructured materials and biological systems. Detailed information of the molecular structure can be obtained, because each vibration of atoms shows a characteristic position and intensity. Both position and intensity are influenced by the chemical bonds, inter- and intramolecular forces [1]

An inelastic scattering process is responsible for the appearance of vibrational bands (Raman spectra). Unlike in IR spectroscopy (where the photon energy must coincide with the energy difference between two available vibrational states), in Raman spectroscopy a molecule can scatter a monochromatic radiation. The Raman spectrum appears in a wavelength λ slightly higher or lower than the incident radiation [1].

Thus Raman scattering can be visualized as a perturbation of the molecule by incident photon, which eigenstates can be described as a linear combination of the stationary states (or time-independent) wave functions of the molecule. The photon released when the molecule returns to electronic ground state has, more often, the same incident photon energy (Rayleigh scattering), but Raman scattering (which contains the vibrational information) can also be present. [1]

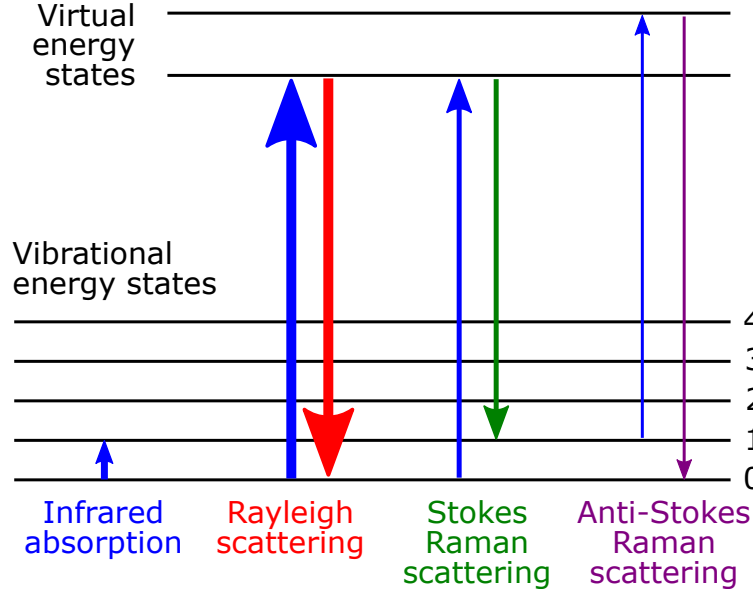


Figure 1: Energy-level diagram showing the states involved in Raman spectra [2]

A vibration (vibration coordinate ρ) is *Raman active* when the polarizability α (or *induced dipolar momentum*) of the molecule is changed (described as $\frac{\partial \alpha}{\partial \rho} \neq \emptyset$). This illustrates that the nature of the Raman effect is physically different from the infrared. Selection rules are also different and thus lead to a different spectra [1].

1.2 Lock-in amplifiers

A lock-in amplifier is a type of amplifier that extracts a signal with a known carrier wave from a noisy environment. Depending on the instrument, signals up to orders of magnitude smaller than the noise components, potentially close by in frequency, can still be reliably detected [3]. This reliability depends on the *dynamic reserve* of the instrument, which is simply a measure of how well the instrument can extract the signal when there's a much larger unwanted noise sitting on top of it. Modern instruments achieve a dynamic reserve up to 120 dB [3].

For amplitude quantities,

$$\text{dB} = 20 \log_{10} \left(\frac{A_{\text{noise}}}{A_{\text{signal}}} \right) \quad (1)$$

So for 120 dB, the above equation yields

$$120 \text{ dB} = 20 \log_{10} \left(\frac{A_{\text{noise}}}{A_{\text{signal}}} \right) \quad (2)$$

which gives

$$\frac{A_{\text{noise}}}{A_{\text{signal}}} = 10^{\frac{120}{20}} = 10^6 \quad (3)$$

So modern instruments can theoretically recover signals with an order of magnitude 10^{-6} smaller than the noise.

A lock-in amplifier performs a multiplication of its input with a reference signal, also sometimes called down-mixing or heterodyne/homodyne detection, and then applies an adjustable low-pass filter to the result [4]. This method is termed demodulation or phase-sensitive detection and isolates the signal at the frequency of interest from all other frequency components. The reference signal is either generated by the lock-in amplifier itself or provided to the lock-in amplifier and

the experiment by an external source. The reference signal is usually a sine wave but could have other forms, too. Demodulation with a pure sine wave enables selective measurement at the fundamental frequency or any of its harmonics [4].

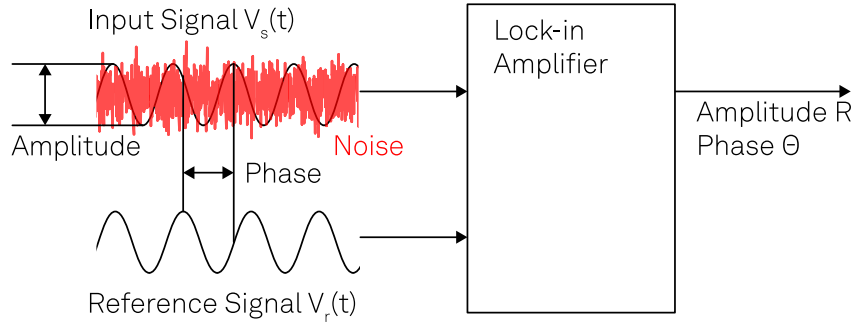


Figure 2: Lock-in amplifiers are capable of measuring the amplitude and the phase of a signal relative to a defined reference signal, even if the signal is entirely buried in noise. Adapted from [4]

1.3 Classical and Quantum Mechanical Theory

The classical picture of Raman scattering starts with the electric field of the incident laser light,

$$E(t) = E_0 \cos(\omega_0 t), \quad (4)$$

which induces a dipole moment in the molecule,

$$p(t) = \alpha E(t), \quad (5)$$

where α is the molecular polarisability.

Since the molecule is vibrating, its polarisability can change with the normal coordinate Q of the vibration:

$$\alpha(Q) = \alpha_0 + \left(\frac{\partial \alpha}{\partial Q} \right)_0 Q. \quad (6)$$

If the vibration has frequency ω_v , the induced dipole contains components at

$$\omega_0, \quad \omega_0 - \omega_v, \quad \omega_0 + \omega_v. \quad (7)$$

These correspond to:

- **Rayleigh scattering:** ω_0
- **Stokes Raman scattering:** $\omega_0 - \omega_v$
- **anti-Stokes Raman scattering:** $\omega_0 + \omega_v$

In the quantum mechanical picture, the incident photon interacts with the molecule through a virtual state. The molecule changes its vibrational energy, and therefore the scattered photon has a different energy:

$$E_{\text{out}} = E_{\text{in}} \mp \hbar\omega_v. \quad (8)$$

Thus, the Raman shift directly provides information about the vibrational energy levels of the molecule.

1.4 Polarisability

Polarisability describes how easily the electron cloud of a molecule can be distorted by an external electric field. The induced dipole moment is given by

$$\mathbf{p}_{\text{ind}} = \boldsymbol{\alpha}\mathbf{E}, \quad (9)$$

where $\boldsymbol{\alpha}$ is generally a tensor.

The important Raman selection rule is

$$\left(\frac{\partial \alpha}{\partial Q} \right)_0 \neq 0 \quad (10)$$

A vibrational mode is therefore **Raman active if the vibration produces a change in the molecular polarisability**.

1.5 Degree of Depolarisation

The degree of depolarisation describes how much the polarisation of the scattered Raman light differs from that of the incident laser light. It is commonly defined as

$$\rho = \frac{I_{\perp}}{I_{\parallel}}, \quad (11)$$

where $I_{\perp}$ is the intensity of the Raman light polarised perpendicular to the incident polarisation and $I_{\parallel}$ is the intensity parallel to it.

For the usual treatment of Raman scattering from randomly oriented molecules,

$$0 \leq \rho \leq \frac{3}{4}. \quad (12)$$

A low value of ρ corresponds to strongly polarised Raman scattering, whereas a larger value indicates stronger depolarisation. The degree of depolarisation can provide information about the symmetry of a vibrational mode.

1.6 Scattering Cross Section

The scattering cross section σ describes the probability that an incident photon will be scattered by a molecule. It can be interpreted as an effective area that characterises the scattering probability.

Raman scattering has a very small scattering cross section compared with Rayleigh scattering. Consequently, Raman signals are weak and require sensitive detectors and sufficiently strong laser illumination.

1.7 Normal Coordinates and Normal Oscillations

A molecule can undergo many different vibrational motions. These motions can be decomposed into independent **normal modes**.

Each normal mode has:

- a characteristic pattern of atomic motion,
- a characteristic vibrational frequency, and
- a normal coordinate Q describing the amplitude of the vibration.

The molecular motion can therefore be described using a set of normal coordinates,

$$Q_1, Q_2, Q_3, \dots \quad (13)$$

which correspond to the different normal modes of vibration.

The dependence of the polarisability on a normal coordinate is particularly important for Raman spectroscopy:

$$\alpha(Q) = \alpha_0 + \left(\frac{\partial \alpha}{\partial Q} \right)_0 Q + \dots \quad (14)$$

1.8 Induced and Permanent Dipole Moment

A molecule may possess a **permanent dipole moment** due to an asymmetric distribution of electric charge,

$$\mathbf{p}_0 \neq 0. \quad (15)$$

An external electric field can also distort the electron cloud and produce an **induced dipole moment**:

$$\mathbf{p}_{\text{ind}} = \alpha \mathbf{E}. \quad (16)$$

Raman scattering originates from this induced dipole. The electric field of the laser drives the induced dipole, which subsequently radiates the scattered light.

1.9 Raman shift wavenumber

Given that energy shifts in the incident photons are not wavelength dependent, it makes sense to express the raman spectra in a wavelength-independent notation. Raman shift wavenumbers are the unit used to do so. The raman shift of a photon is simply

$$\Delta\nu = \nu_0 - \nu_s \quad (17)$$

where ν_0 is the frequency of the incident light and ν_s is the frequency of the scattered light.

This way we can express raman shifts without involving laser wavelengths knowing well that raman spectra will show in the same wavenumber regardless of the laser used.

2 Methodology

2.1 Calibration of the Monochromator

First the monochromator's transmission curves for horizontally and vertically polarized light were looked after. To do so, a Quartz Tungsten-Halogen Lamp was used. This lamp features a stable output and broadband, incoherent illumination in the visible and near-IR portions of the spectrum.

A polarizer was placed at the input of the monochromator as show in figure 3. For further analysis, it is important that both transmission curves are comparable, and hence measured with the same set parameters in the experiment (or somehow made compatible through data processing). After tweaking the parameters of the coupler and the photomultiplier high voltage supply, the following configuration was found to have enough dynamic range to record both horizontal and vertical without saturating the instrument signals: The setup was first manipulated as to fit the output voltage obtained by horizontally polarized light best, just below the saturation point for the whole spectra. The coupler was set in 10^{-6} , the voltage supply for the photomultiplier to ~ 300 V and the spectra was taken from 400 nm to 800 nm at a speed of 5 nm/s.

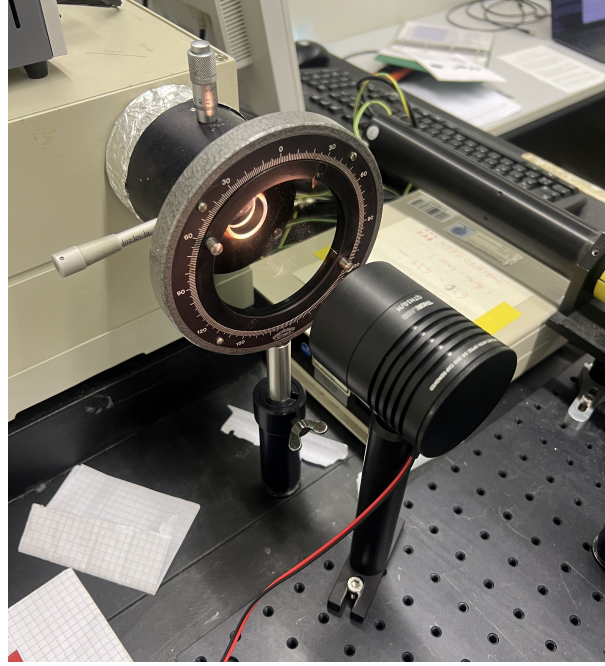


Figure 3: Halogen lamp, polarizer and monochromator input

Then, with the same set of parameters, the polarizer was vertically aligned and once again the spectra was taken from 400 nm to 800 nm at the same speed.

The curve corresponding to horizontal polarization can be seen in figure 4. The curve corresponding to vertical polarization can be seen in figure 5. Both of these figures are raw images extracted from the TDS2001IC oscilloscope screen.

In the experiment, the monochromator grating is adjusted by a servo motor connected to a microcontroller which was programmed to make precise adjustments to the grating. In order to know which voltage corresponds to which wavelength, the arduino's voltage output is also recorder in the oscilloscope to match the data.

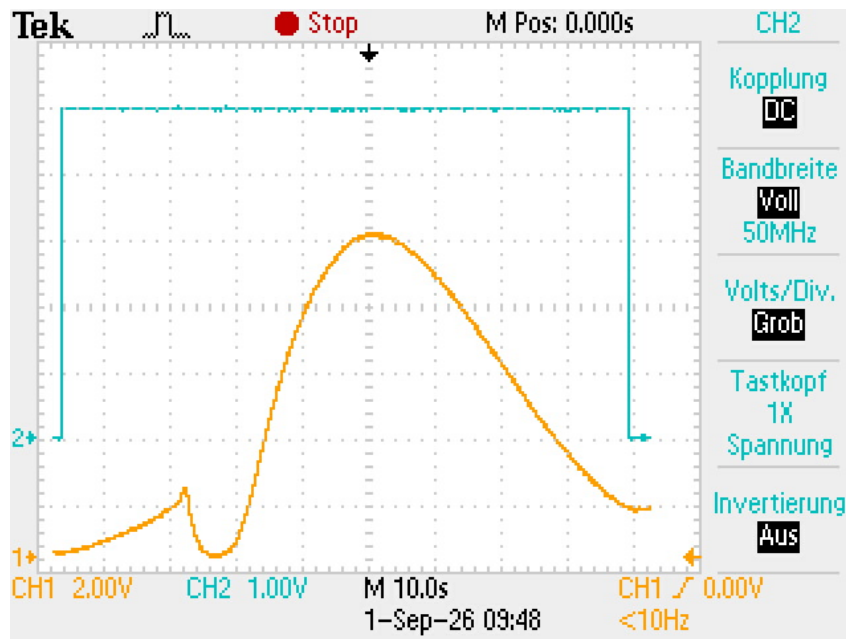


Figure 4: Voltage curve of the photomultiplier for horizontally polarized light (orange) and servo voltage (blue).

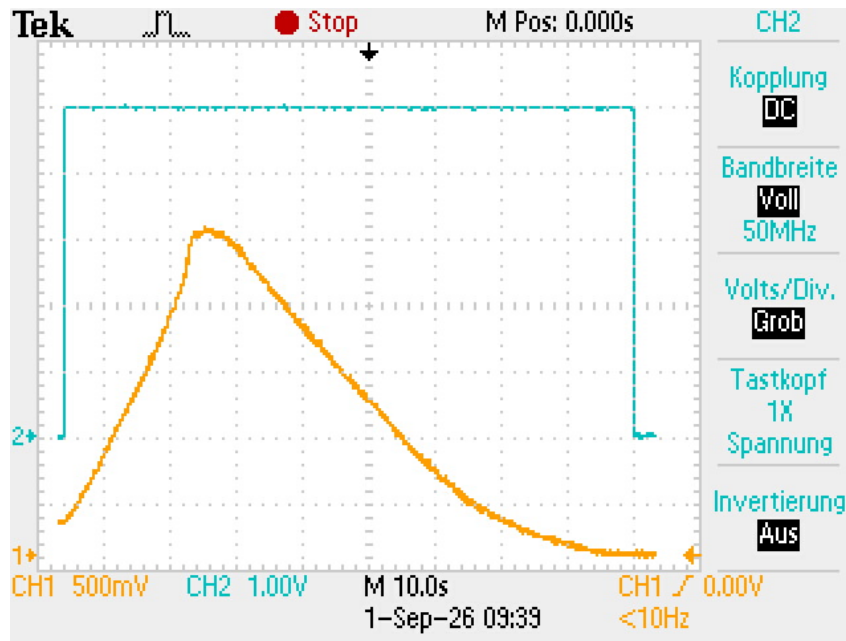


Figure 5: Voltage curve of the photomultiplier for vertically polarized light (orange) and servo voltage (blue).

The data shown in both figures 4 and 5 will be useful to apply instrument response correction (correct the actual raman spectra taking into account the wavelength-dependent intensity of the photocromator).

The Halogen lamp was removed from the array, and the two lenses that form the telescope were placed so that the laser light goes in as focused as possible in the monochromator. The ethanol + salt sample was inserter in the chamber. This sample has a very strong rayleigh reflection, and is thus the ideal sample to align the experiment. The monochromator was swept in between 622 and 642 nm. The Voltage curve can be seen in figure 6. Note that the lock-in mechanism and the chopper are not active yet.

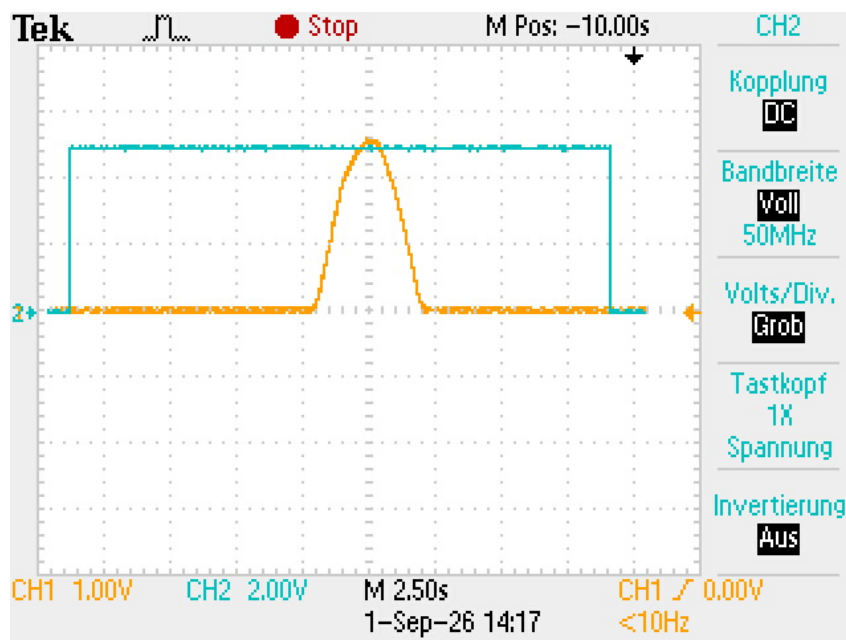


Figure 6: Voltage curve of the 633 nm laser (ethanol +salt cuvette)

Figure 6 shows that our laser indeed has a gaussian profile with a peak presumably at 633 nm.

2.1.1 instrument response correction

Figures 4 and 5 prove that the monochromator has a wavelength-dependent response $R(\lambda)$, thus we can say that approximately

$$I_{measured}(\lambda) = I_{true}(\lambda)R(\lambda) \quad (18)$$

So to correct the measured spectrum we divide by the response of the monochromator.

2.2 Experiment Setup

Having the monochromator response, the rest of the setup can be properly introduced in the experiment. A laser of wavelength of 633 nm goes through a mechanical chopper whose frequency can be adjusted by a control box. The laser then goes through a cage system containing a sample, it is then reflected onto the telescope system. Finally the laser goes inside the monochromator. the photomultiplier then outputs a signal for data analysis. In between the sample and the monochromator, a polarizer is introduced to analyze the change in light polarization after the sample. The laser polarization can also be adjusted by rotating the laser tube.

A diagram of the experiment can be seen in 7.

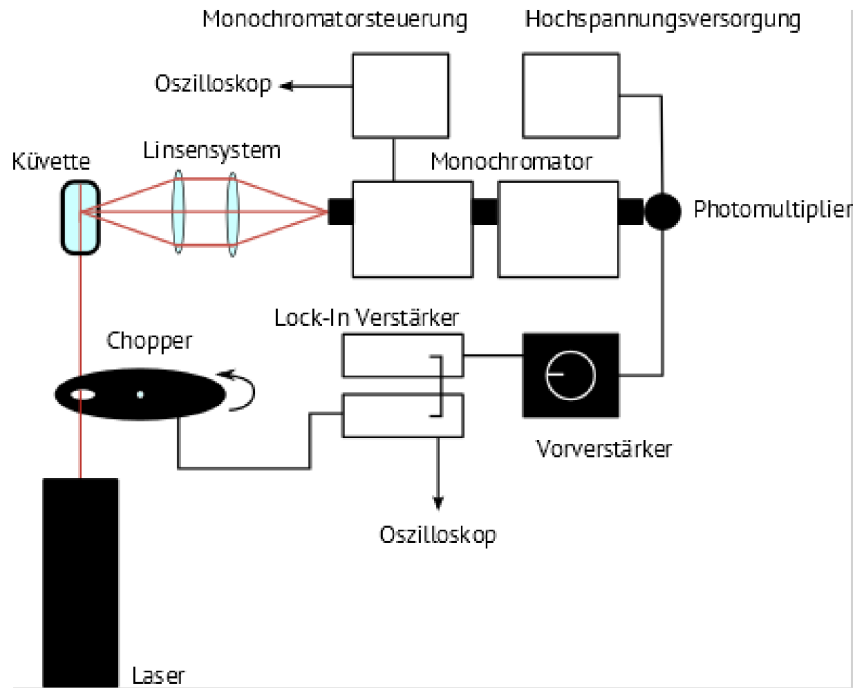


Figure 7: Setup of the experiment. Taken from [5]

After the monochromator, the signal processing electronics take place. Given that the photomultiplier signal is very small, immediately after a preamp is connected to strengthen the signal. Said signal is then fed to a phase sensitive Detector. This detector has as reference the phase-shifted signal of the chopper. A picture of this part of the array is seen in figure 8. These electronics and configuration is explained in detail in section 2.3.1

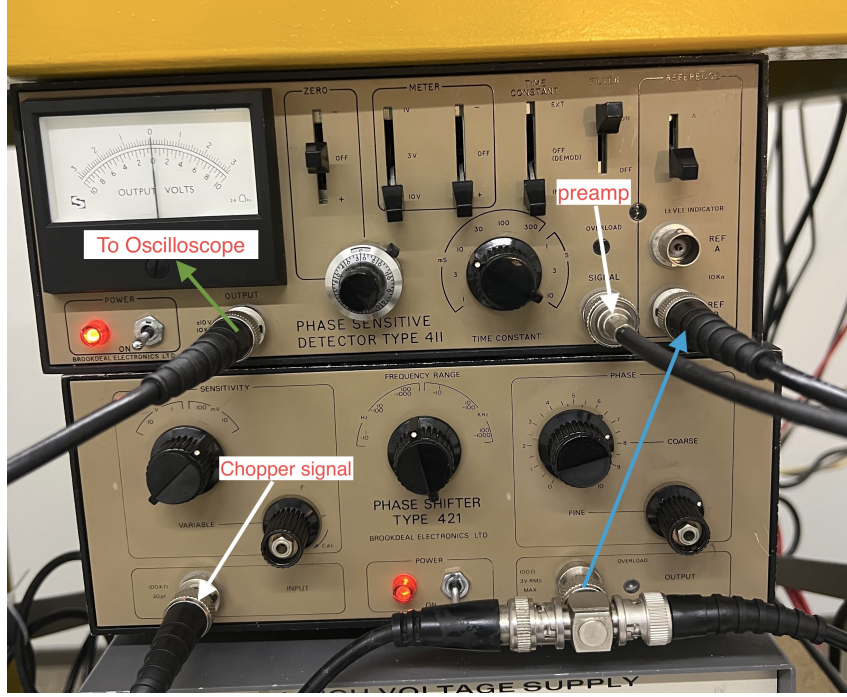


Figure 8: Lock-In amplifier inputs and outputs (knobs are not in final position)

2.3 Lock-In Amplifier

It will be seen in the following sections of this protocol that raman peaks are of very low intensity and can be hard to discern in a regular spectrum; this is because raman scattering is considerably weaker than rayleigh scattering. Often they can be buried in noise and the rayleigh peaks can also bury them if they're overlapping.

To overcome this challenge, we need a way to remove unwanted noise from the spectra. This is where the Lock-In amplifier comes in handy as explained in section 1.2

2.3.1 Lock-in amplifier configuration

The optical chopper was set to an arbitrary frequency (around 130 Hz). The chopper driver generates an AC output that was forwarded to our Phase shifter. The output of the phase shifter is then taken as a reference signal in the phase sensitive detector (PSD). Furthermore, the chopper frequency can be calculated with the AC output signal. The output of the current amplifier (whose input is the photomultiplier) is then taken as the input signal on the PSD. Refer to figure 8 for a visual image of the connections.

The fine-tuning of the lock-in mechanism took place. Given that the chopper has a frequency of around 130 Hz, the phase shifter frequency has to be set accordingly to the range 100 – 1000 Hz. Then the phase shift is adjusted either to match closely either 0° or 180°. This is because the photodiode signal at the chopping frequency has the form

$$S(t) = A \cos(\omega t + \phi) \quad (19)$$

while the reference from the chopper is

$$R(t) = \cos(\omega t + \theta) \quad (20)$$

At the PSD, both signals are multiplied and low-pass filtered, giving an expression resembling

$$V(t) = \frac{A}{2} \cos(\phi - \theta) \quad (21)$$

It is evident that the resulting signal has an absolute maxima when the above condition is met. In practice, this is done by hooking the photodiode and the phase-shifted chopper in the two channels of the oscilloscope and adjusting the time scale to distinguish the square oscillating signal as seen in figure 9

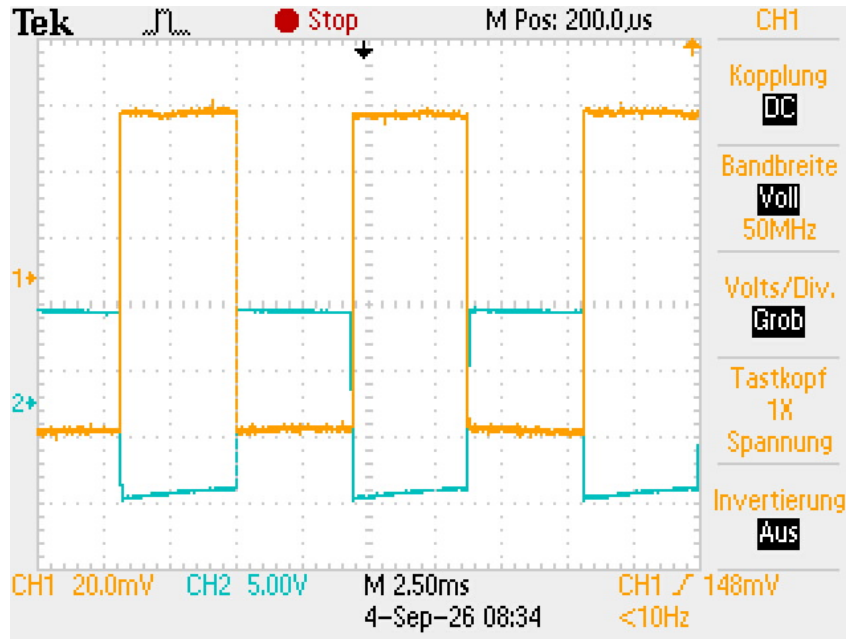


Figure 9: Phase shifted chopper (blue) and photodiode (yellow) on the oscilloscope. They fulfill the locked condition

The time constant α of the PSD was introduced after that. The time constant is the low pass constant which determines how quickly the PSD output is allowed to respond to changes by controlling the bandwidth of its low pass filter.

Simply put, if α is small, the output responds quickly, but more noise gets through. Viceversa, if α is big, output changes slowly, but more noise is rejected.

3 Results and Discussion

After processing the data by taking into account the servo response, both of the curves seen in figures 4 and 5 are now shown with the wavelength (nm) in figures 10 and 11

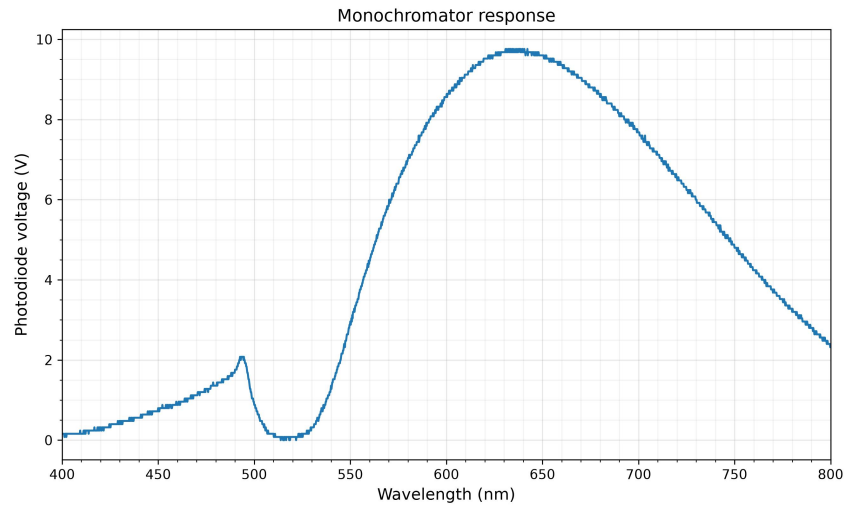


Figure 10: Monochromator response (horizontally polarized light).

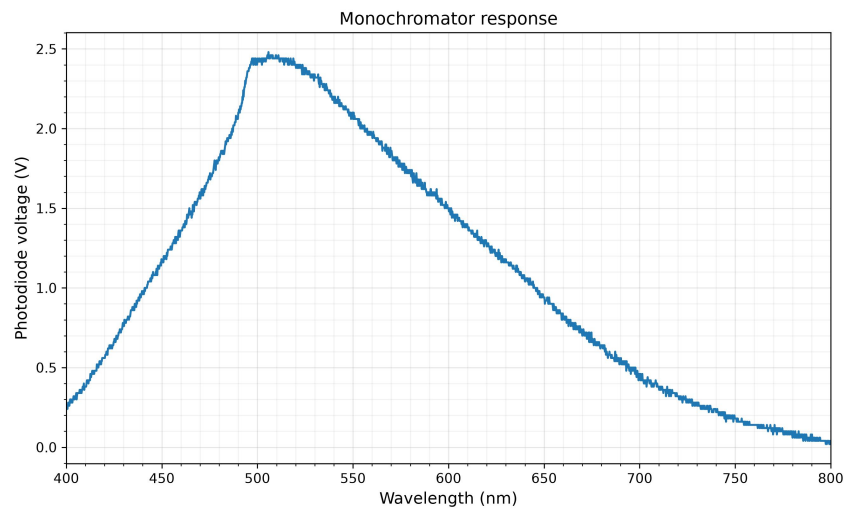


Figure 11: Monochromator response (vertically polarized light).

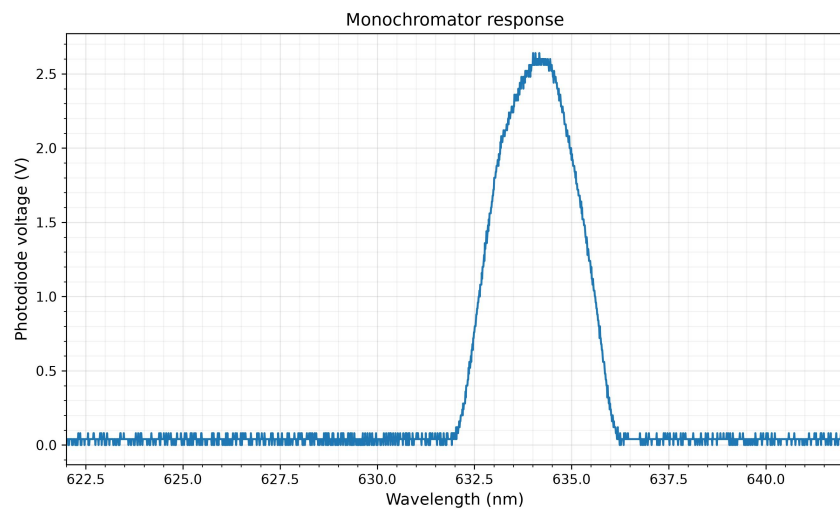


Figure 12: Laser spectra (ethanol + salt)

Take note that all of the next results are corrected to consider the monochromator polarizability/wavelength response as explained in section 2.1.1.

3.1 Spectra and fitting of CCL4

As suggested in section 1.9, all of the results are presented in the appropriate raman shift wavenumber.

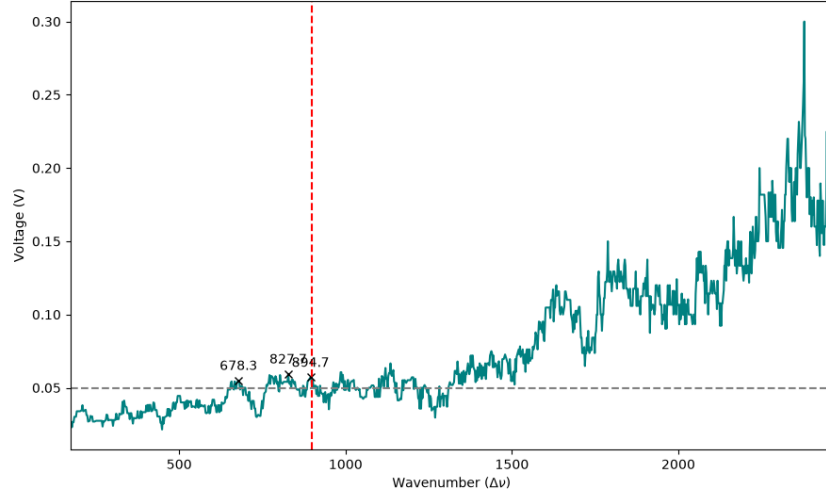


Figure 13: CCL4 spectra with laser at 0° and polarizer at 0°

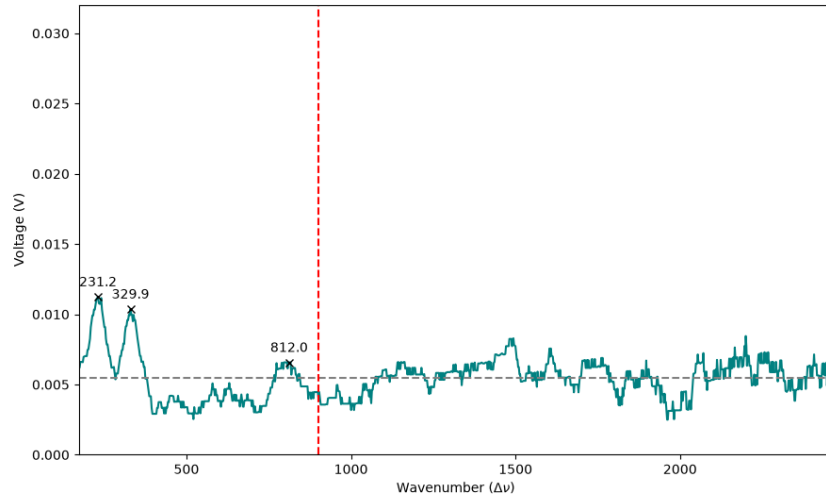


Figure 14: CCL4 spectra with laser at 0° and polarizer at 90°

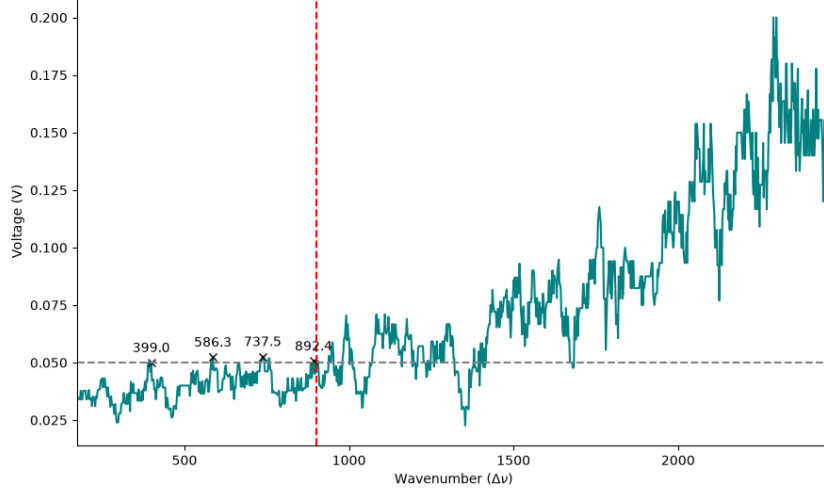


Figure 15: CCL4 spectra with laser at 90° and polarizer at 0°

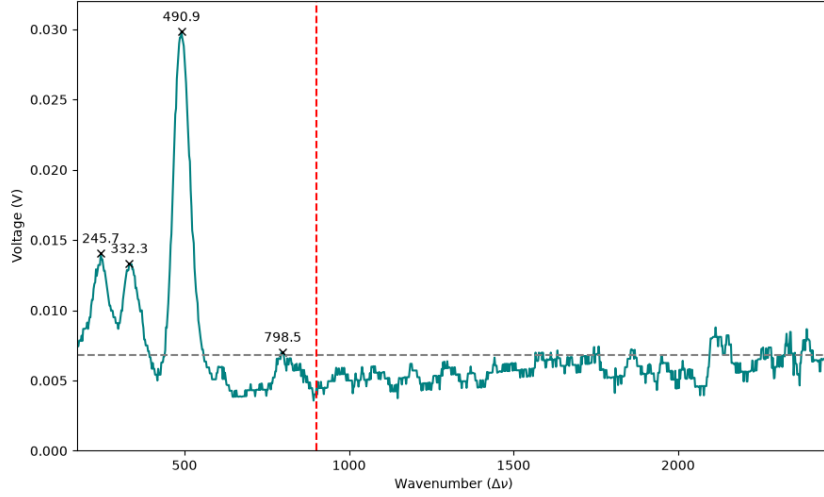


Figure 16: CCL4 spectra with laser at 90° and polarizer at 90°

Taking a closer look at the CCL4 spectra above, notice that not all of the laser/analyzer configurations yield expected results. If we consider exclusively the sample and the relative orientation of the incident polarization and analyzer, we would expect a symmetrical behavior of the experiment configurations. CCL4 has totally symmetric A modes and non-symmetric F modes.

If we simplify the experiment to two orthogonal polarization directions x and y , we can think of the experiment yielding symmetric results as in table 1. This is however not the case. In reality, the experiment doesn't yield the expected symmetry but instead the ones shown in table 2.

Given that CCL4 is an isotropic sample, there is no molecular reason for this to disappear while the other survives. The most likely explanation is that the optical system has an additional polarization-dependent transmission/collection effect. As already mentioned, the monochromator used in the experiment has a different response to the orientation of light. This combined with an improper optical alignment is one likely explanation as to why we cannot resolve the lines with the analyzer at the 0° degrees configuration.

Table 1: Idealized polarization dependence of Raman scattering in an isotropic CCl_4 sample. The A_1 mode produces polarized scattering, while the F_2 modes produce depolarized scattering.

Laser polarization	Analyzer polarization	
	0°	90°
0°	A_1	F_2
90°	F_2	A_1

Table 2: Results of the experiment

Laser polarization	Analyzer polarization	
	0°	90°
0°	noise	F_2
90°	noise	A_1

To discuss the number of frequency shifts in our spectra, it's relevant to explain the molecular structure of CCl_4 and its possible vibrational modes. A very didactic animation can be found on [6]([click here](#)). A diagram is also sketched for those who are not reading this report on an e-device.

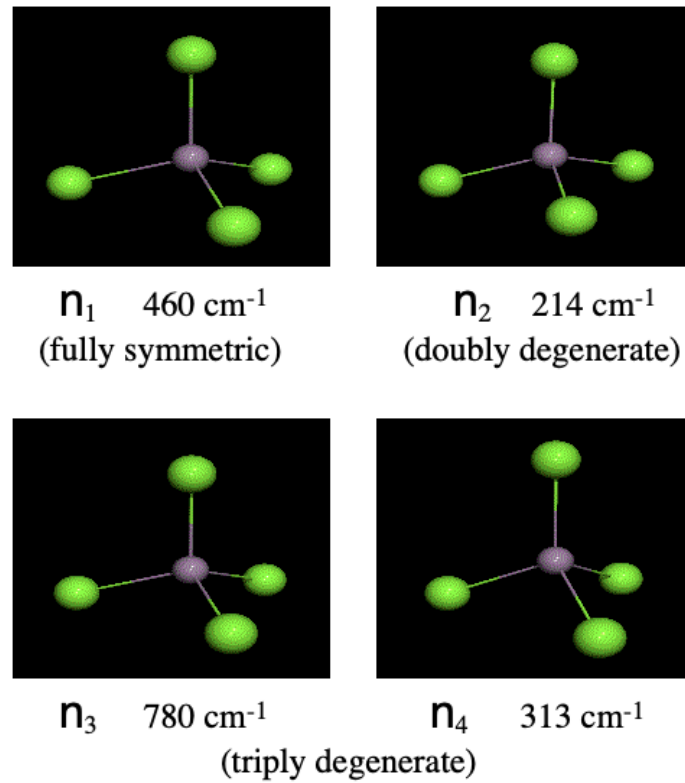


Figure 17: Vibrational modes of Carbon tetrachloride. Adapted from [6]

Each of these vibrational modes influences the amount of lines seen in the raman spectra.

The vibrational modes of CCl_4 can be classified according to the symmetry of the molecule, resulting in four distinct fundamental modes: $\nu_1(A_1)$, $\nu_2(E)$, $\nu_3(F_2)$, and $\nu_4(F_2)$. Consequently, four distinct fundamental vibrational frequencies are expected in the Raman spectrum, and are indeed seen in figure 16, although the peak corresponding to ν_3 is barely perceptible in our measurements.

Peak ν_3 corresponds to a fermi resonance. Because of this, instead of observing a single peak, we see a double peak profile in figure 18. This is why the fit shown in figure 21 accounts for a two peak fit, even if in our data this distinction is not perceived.

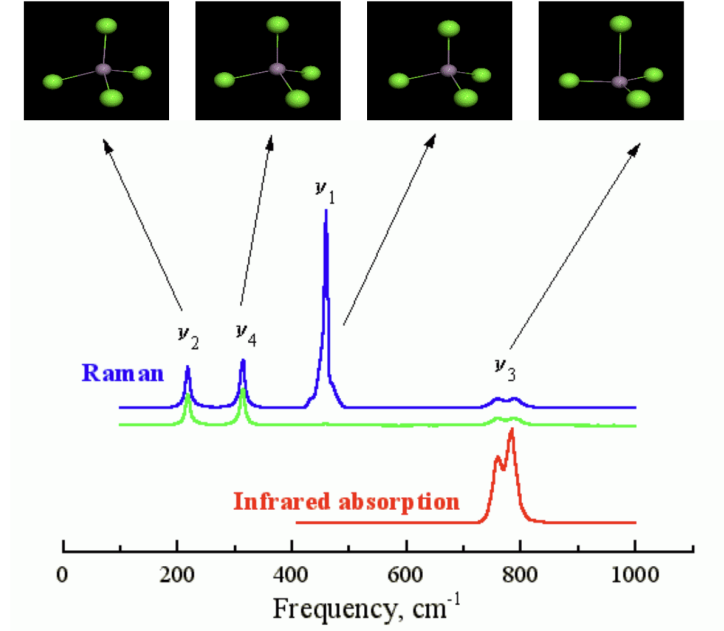


Figure 18: Raman peaks corresponding to vibrational modes. Adapted from [6]

The following curve fits (figures 19, 20 and 21) were done using the spectra from the measurement shown in figure 16.

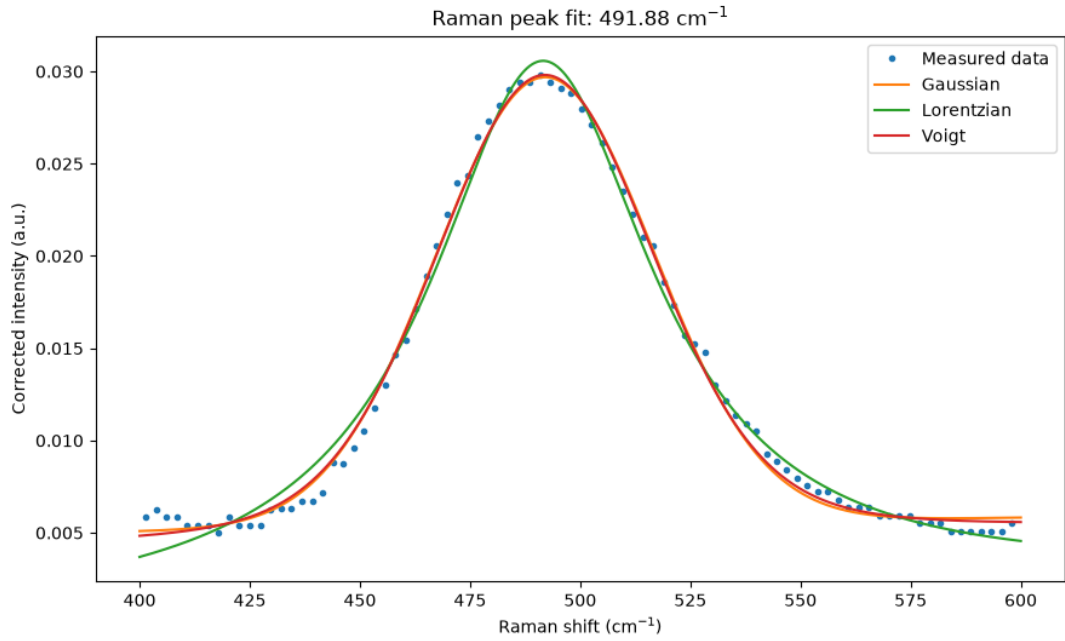


Figure 19: Fit for Raman peak at 491.9 cm^{-1}

Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
Gaussian	492.0 ± 0.2	653.35 ± 0.01	58.2	0.9954
Lorentzian	491.4 ± 0.3	653.32 ± 0.01	61.8	0.9902
Voigt	491.9 ± 0.2	653.34 ± 0.01	58.8	0.9956

Table 3: Fit parameters for the 491.9 cm⁻¹ Raman peak.

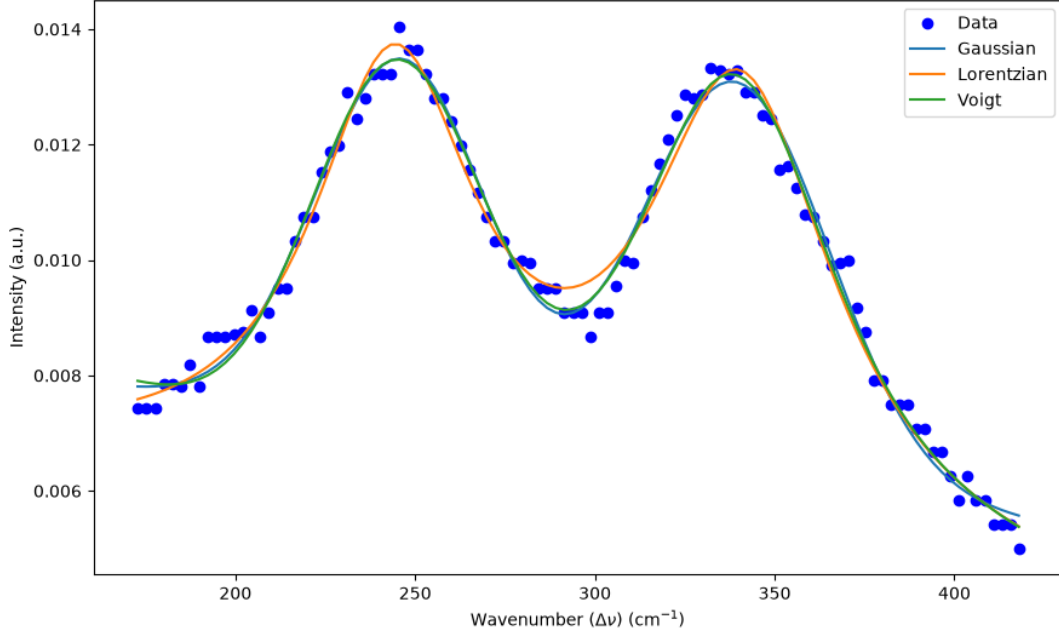


Figure 20: Fit for the peaks ~ 245 cm⁻¹ and ~ 332 cm⁻¹.

Fit	$\Delta\nu$ (cm ⁻¹)	FWHM (cm ⁻¹)	R^2
Gaussian	246.2 ± 0.5	56.0	0.9838
Gaussian	339.0 ± 0.4	62.0	0.9838
Lorentzian	244.2 ± 0.5	56.1	0.9806
Lorentzian	340.5 ± 0.5	64.4	0.9806
Voigt	245.0 ± 0.6	52.4	0.9848
Voigt	338.7 ± 0.8	67.5	0.9848

Table 4: Fit parameters for the twin peaks ~ 245 cm⁻¹ and ~ 332 cm⁻¹ Raman peak.

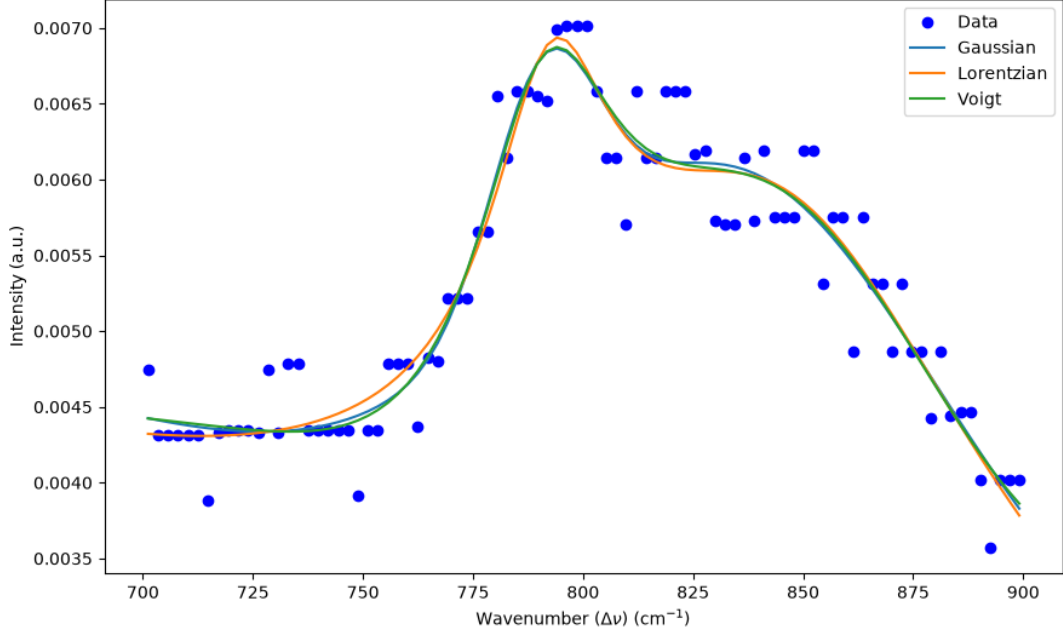


Figure 21: Fit for fermi resonance near $\sim 800 \text{ cm}^{-1}$

Fit	$\Delta\nu \text{ (cm}^{-1}\text{)}$	FWHM (cm $^{-1}$)	R^2
Gaussian	791.1 ± 1.2	28.3	0.9212
Gaussian	835.5 ± 4.5	105.5	0.9212
Lorentzian	792.9 ± 1.2	33.6	0.9179
Lorentzian	845.1 ± 4.4	121.5	0.9179
Voigt	791.6 ± 1.4	41.6	0.9215
Voigt	843.5 ± 4.6	65.7	0.9215

Table 5: Fit parameters for the twin peaks 245 cm^{-1} and 332 cm^{-1} Raman peak.

The intrinsic line shape of a Raman transition is expected to be Lorentzian when the vibrational coherence decays exponentially due to finite lifetime. In an experimental spectrum, however, additional instrumental and inhomogeneous broadening can modify this line shape, resulting in Gaussian or Voigt profiles. For the fits seen in figures 19, 20 and 21, the best approximation happens to be a Voigt profile, which corresponds to the convolution of the gaussian response of the instrument and the lorentzian raman profiles. This is a likely explanation as to why the R^2 value of the fits is maximized using a Voigt profile.

3.1.1 literature comparison of raman peaks: CCL4

For tables 6 and 8, the following notation is useful: $C = 3 \sim 6 \text{ cm}^{-1}$, $E = 15 \sim 30 \text{ cm}^{-1}$,

As it is shown on table 6, our measurements are off by a few dozens of wavenumbers. Most likely the reference between the software/monochromator was not properly adjusted before the measurments, and hence a displacement of wavenumbers occured in our data.

The reported values in column *Measured ν* in Table 6 takes into account the fits done previously.

Table 6: Literature and measured Raman bands of carbon tetrachloride (CCl_4).

Symmetry	Mode	NIST ν 1/cm	Uncertainty 1/cm	Measured ν 1/cm	Deviation 1/cm
a_1	Symmetric stretch	458.7	C	491.9	32.2
e	Degenerate deformation	217.0	C	245.0	28.7
f_2	Degenerate stretch	790.4	E	843.5	53.1
f_2	Degenerate stretch	761.7	E	791.6	29.9
f_2	Degenerate deformation	313.5	C	338.7	18.8

3.1.2 polarization of CCL4 modes

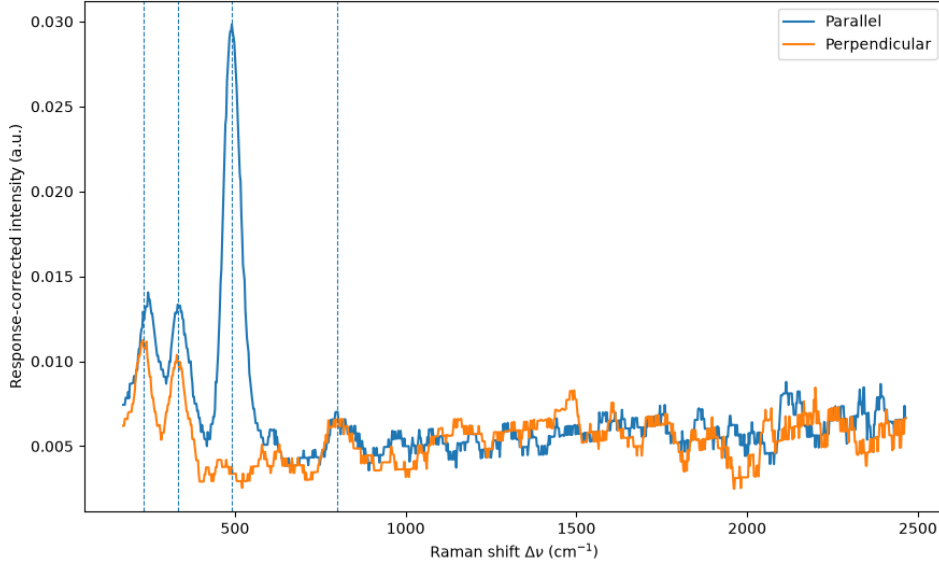


Figure 22: Horizontal and Perpendicular measurements superimposed

$\Delta\nu$ (cm^{-1})	$I_{\parallel}$ (a.u.)	$I_{\perp}$ (a.u.)	ρ
231	0.0132	0.0112	0.851
332	0.0133	0.0104	0.778
490	0.0298	0.00378	0.127
798	0.00702	0.00658	0.937

Table 7: polarization ratios for the vibrational modes of CCL4

Table 7 shows the calculated polarization ratios from the measurements taken. Our measurements are in agreement with the literature reported in [7]. The only strongly polarized peak corresponds to the vibrational mode $\nu(A_1)$, since it corresponds to a diagonal entry in the raman polarization tensor of CCl_4 . The rest of the peaks ν_2 , ν_3 and ν_3 correspond to non-diagonal entries in the tensor, and are thus depolarized as seen in both figure 22 and table 7

3.2 Spectra and fitting of Benzene

Due to technical issues during the original measurement, the data acquired during the experiment could not be used for the subsequent analysis. Therefore, the analysis was performed using

experimental data provided by the experiment supervisor Dennis Leonard, which were acquired under the corresponding experimental conditions.

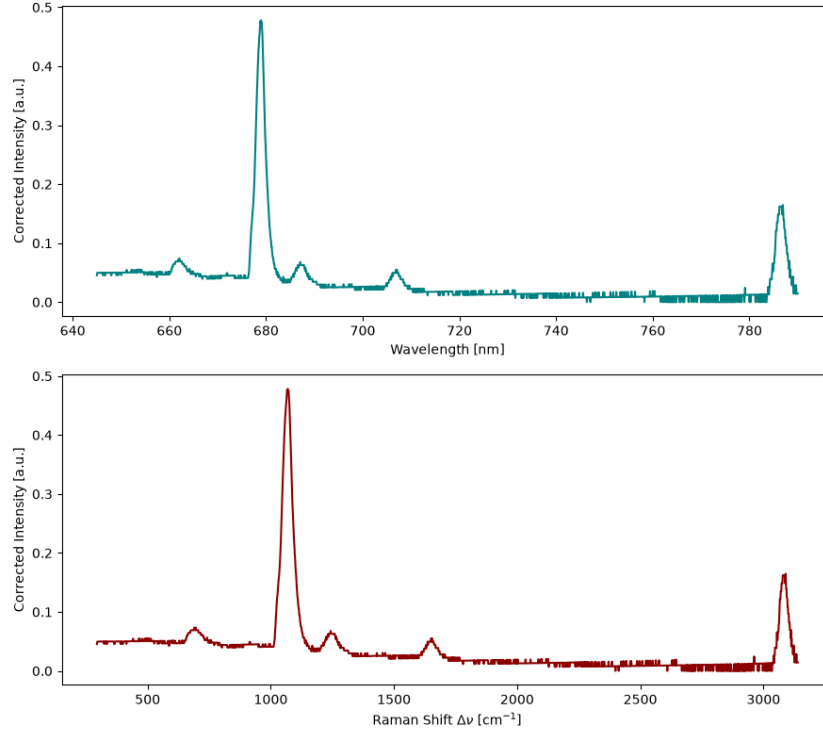


Figure 23: Stokes spectra of Benzene. Data provided by Dennis Leonard.

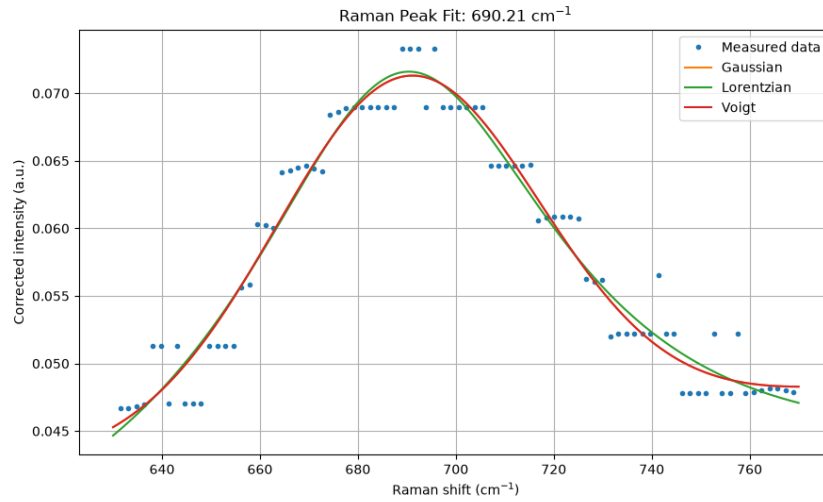


Figure 24: Fit for peak $\sim 690.21 \text{ cm}^{-1}$

Fit	$\Delta\nu \text{ (cm}^{-1}\text{)}$	$\lambda \text{ (nm)}$	FWHM (cm^{-1})	R^2
Gaussian	690.2 ± 0.8	661.92 ± 0.04	63.3	0.9539
Lorentzian	689.4 ± 0.9	661.88 ± 0.04	80.6	0.9511
Voigt	690.2 ± 0.9	661.92 ± 0.04	63.3	0.9539

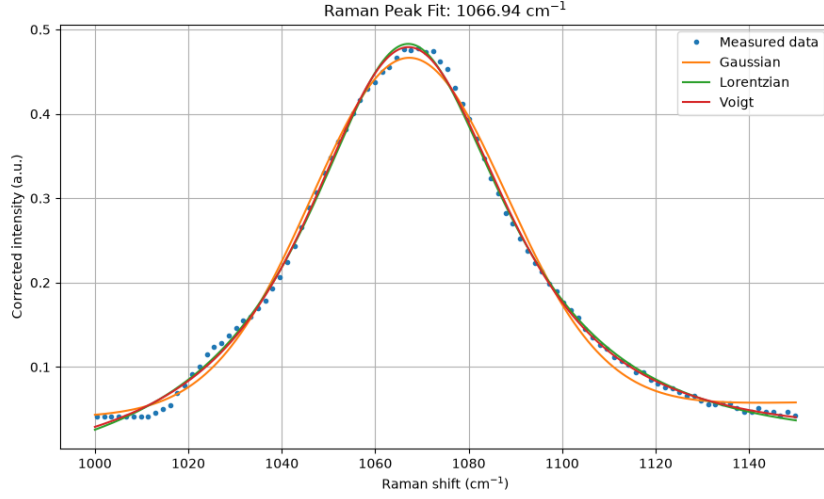


Figure 25: Fit for peak at $\sim 1066.94 \text{ cm}^{-1}$

Fit	$\Delta\nu \text{ (cm}^{-1}\text{)}$	$\lambda \text{ (nm)}$	FWHM (cm^{-1})	R^2
Gaussian	1067.2 ± 0.2	678.86 ± 0.01	49.2	0.9944
Lorentzian	1066.9 ± 0.1	678.85 ± 0.01	53.3	0.9978
Voigt	1066.9 ± 0.1	678.85 ± 0.01	52.2	0.9981

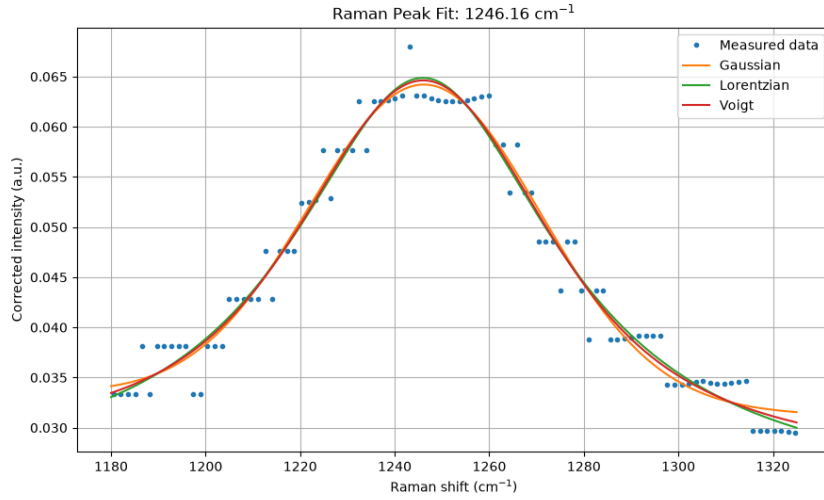


Figure 26: Fit for peak at $\sim 1246.16 \text{ cm}^{-1}$

Fit	$\Delta\nu \text{ (cm}^{-1}\text{)}$	$\lambda \text{ (nm)}$	FWHM (cm^{-1})	R^2
Gaussian	1246.4 ± 0.5	687.22 ± 0.03	57.6	0.9692
Lorentzian	1246.1 ± 0.6	687.20 ± 0.03	68.8	0.9703
Voigt	1246.2 ± 0.6	687.21 ± 0.03	64.1	0.9709

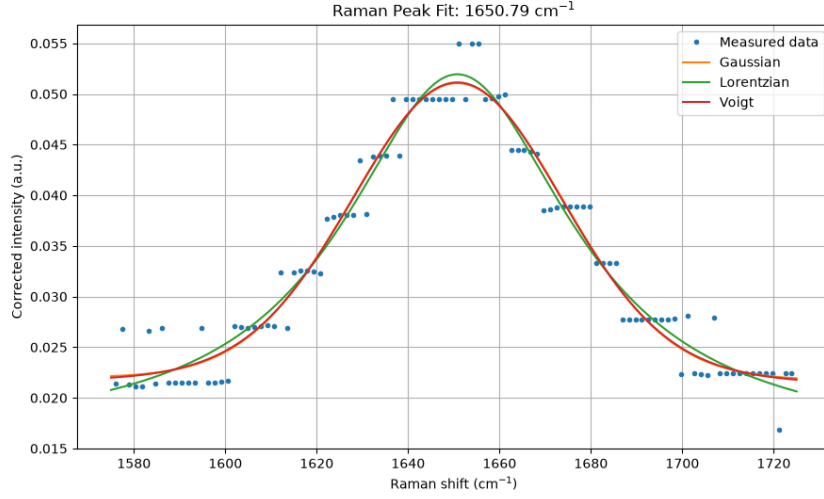


Figure 27: Fit for peak at $\sim 1650 \text{ cm}^{-1}$

Fit	$\Delta\nu \text{ (cm}^{-1}\text{)}$	$\lambda \text{ (nm)}$	FWHM (cm^{-1})	R^2
Gaussian	1650.8 ± 0.5	706.86 ± 0.03	54.2	0.9605
Lorentzian	1650.8 ± 0.6	706.86 ± 0.03	61.2	0.9587
Voigt	1650.8 ± 0.5	706.86 ± 0.03	54.9	0.9605

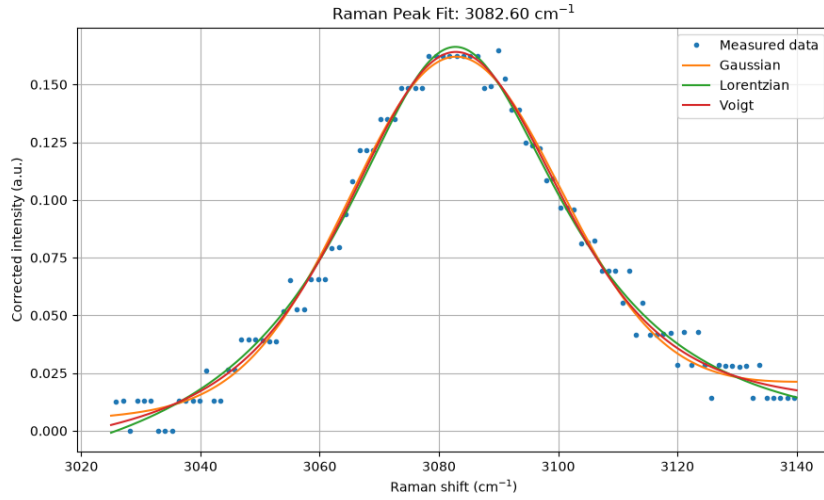


Figure 28: Fit for peak at $\sim 3082 \text{ cm}^{-1}$

Fit	$\Delta\nu \text{ (cm}^{-1}\text{)}$	$\lambda \text{ (nm)}$	FWHM (cm^{-1})	R^2
Gaussian	3082.6 ± 0.3	786.46 ± 0.02	41.4	0.9847
Lorentzian	3082.6 ± 0.3	786.46 ± 0.02	47.0	0.9846
Voigt	3082.6 ± 0.3	786.46 ± 0.02	44.0	0.9857

3.2.1 Benzene antistokes

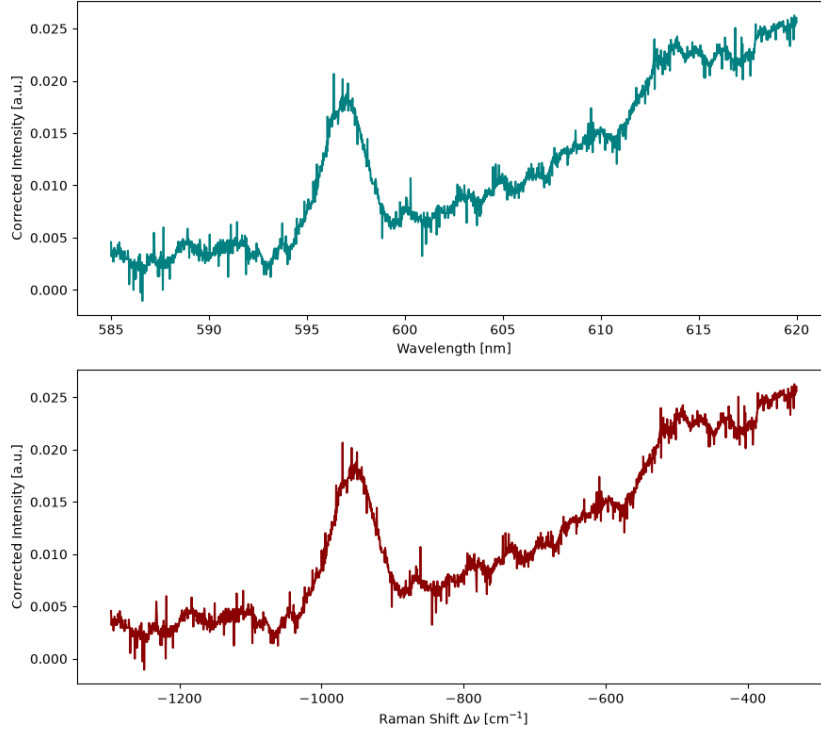


Figure 29: Antistokes spectrum of Benzene. Data provided by Dennis Leonard

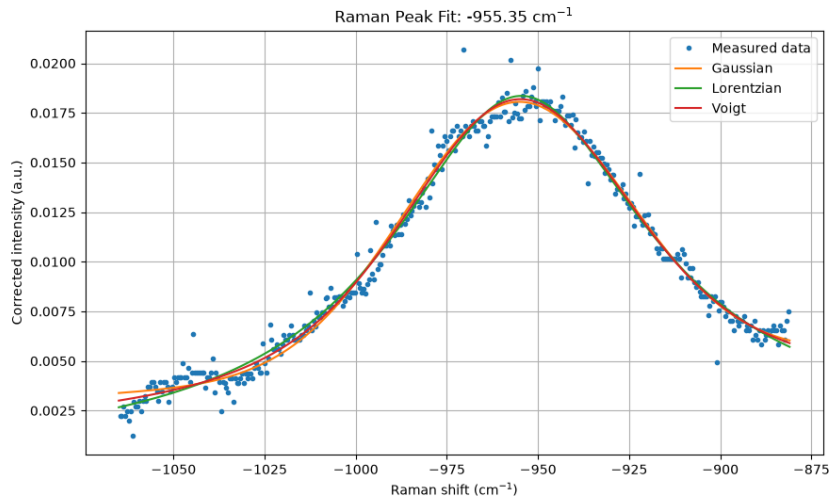


Figure 30: Fit for peak at $\sim -955 \text{ cm}^{-1}$

Fit	$\Delta\nu \text{ (cm}^{-1}\text{)}$	$\lambda \text{ (nm)}$	FWHM (cm^{-1})	R^2
Gaussian	-955.9 ± 0.3	596.89 ± 0.01	73.0	0.9826
Lorentzian	-954.8 ± 0.3	596.92 ± 0.01	87.1	0.9824
Voigt	-955.4 ± 0.3	596.90 ± 0.01	79.0	0.9835

The spectrum seen in figure 29 shows only the strongest of the peaks seen in the stokes regime. This is likely because at room temperature, most of the molecules are at the lowest vibrational

state, and thus little anti-stokes transitions can occur.

The comparison between the reported values of Benzene transitions and the measured ones is shown in table 8.

Symmetry	Mode	NIST ν 1/cm	Uncertainty 1/cm	Measured ν 1/cm	Deviation 1/cm
a_{1g}	CH stretch	3061.9	C	3082.0	20.1
e_{2g}	Ring stretch	1606.4	E	1650.8	44.4
e_{2g}	Ring stretch	1584.6	E	—	—
a_{1g}	Ring stretch	991.6	C	—	—
e_{2g}	CH bend	1178.0	C	—	—
e_{1g}	CH bend	848.9	C	—	—
e_{2g}	Ring deformation	605.6	C	—	—
—	Combination/overtone	~703	—	690.2	~12.8
—	Unassigned feature	—	—	1067.0	—
—	Unassigned feature	—	—	1246.0	—

Table 8: Literature and measured Raman bands of benzene (C_6H_6). The assignments of the measured bands are based on their proximity to literature values; features without a close correspondence to a fundamental vibrational mode are listed as tentative combination/overtone features.

The measured Raman spectrum of benzene exhibits peaks at 690.21, 1067, 1246, 1651, and 3082 cm^{-1} . Comparison with literature values shows that the observed features can be associated with several vibrational motions of the benzene molecule, although some assignments are less certain. The peak at 690.21 cm^{-1} lies in the region of benzene ring-deformation modes and may be related to an overtone or combination band; a calculated combination feature has been reported near 703 cm^{-1} . The peaks at 1067 and 1246 cm^{-1} do not correspond closely to the principal fundamental Raman-active modes of benzene and may therefore originate from combination or overtone bands, or from experimental effects. The feature at 1651 cm^{-1} occurs in the region of the e_{2g} ring-stretching vibration, for which Raman lines are reported around 1585–1606 cm^{-1} ; this region is also affected by Fermi resonance between vibrational states. Finally, the peak at 3082 cm^{-1} lies in the C–H stretching region and can be associated with the Raman-active C–H stretching modes of benzene, reported around 3047–3062 cm^{-1} . The systematic shift between several measured and literature frequencies should be considered when interpreting these assignments, as it may indicate an offset or calibration error in the Raman-shift determination.

3.3 Unknown sample Spectra

Figure 31 shows the spectra recorded for the sample whose content was not known in advance.

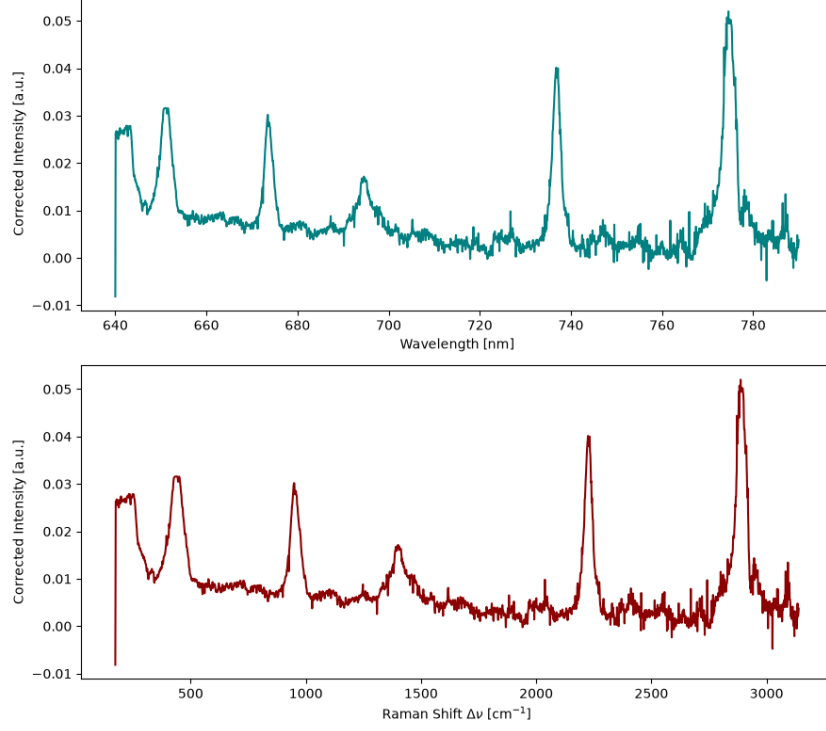


Figure 31: Spectra of *unknown sample 1*. Data provided by Dennis Leonard

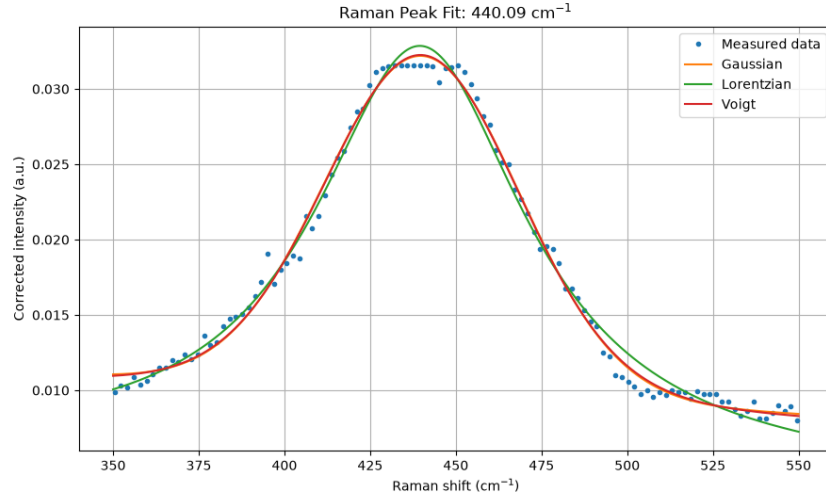


Figure 32: Spectra of *peak at 450 cm⁻¹*.

Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
Gaussian	440.1 ± 0.3	651.14 ± 0.01	67.2	0.9929
Lorentzian	439.7 ± 0.4	651.12 ± 0.02	74.9	0.9889
Voigt	440.1 ± 0.3	651.14 ± 0.01	67.9	0.9930

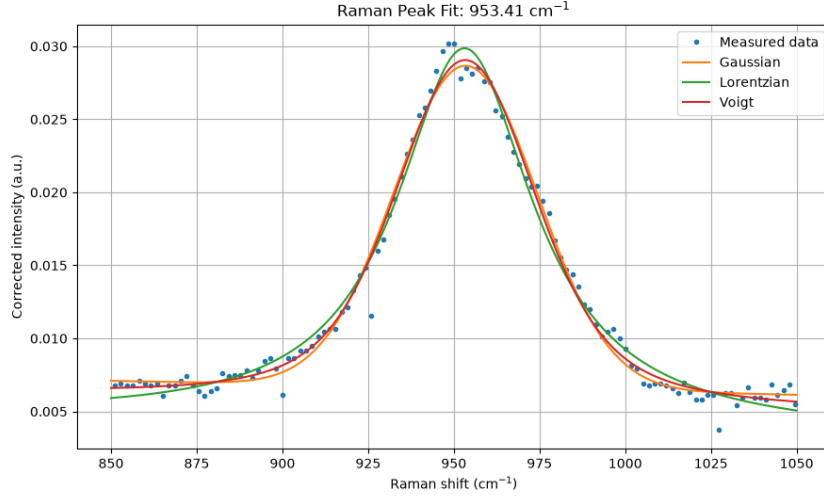


Figure 33: Fit for *peak at* 950 cm^{-1} .

Fit	$\Delta\nu\text{ (cm}^{-1}\text{)}$	$\lambda\text{ (nm)}$	FWHM (cm^{-1})	R^2
Gaussian	953.5 ± 0.3	673.66 ± 0.01	49.2	0.9876
Lorentzian	953.2 ± 0.3	673.64 ± 0.01	48.5	0.9852
Voigt	953.4 ± 0.2	673.66 ± 0.01	49.0	0.9889

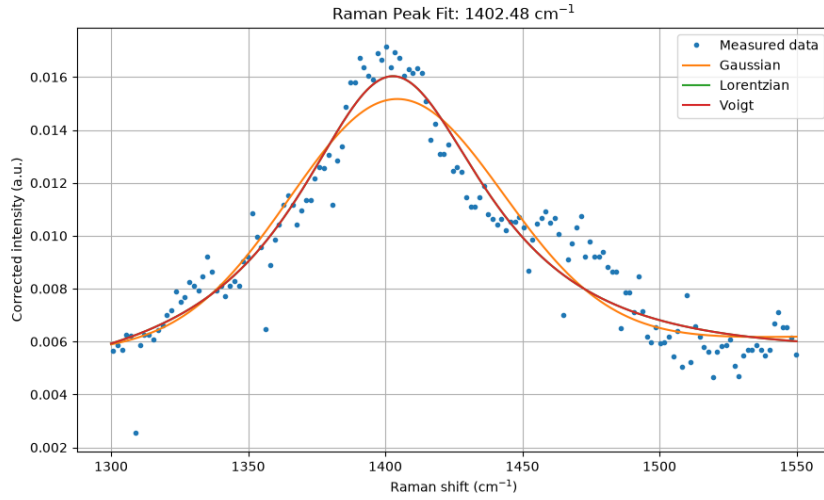


Figure 34: Fit for *peak at* 1410 cm^{-1} .

Fit	$\Delta\nu\text{ (cm}^{-1}\text{)}$	$\lambda\text{ (nm)}$	FWHM (cm^{-1})	R^2
Gaussian	1404.0 ± 1.3	694.75 ± 0.06	91.8	0.8903
Lorentzian	1402.5 ± 1.0	694.67 ± 0.05	86.8	0.9146
Voigt	1402.5 ± 1.0	694.67 ± 0.05	86.8	0.9146

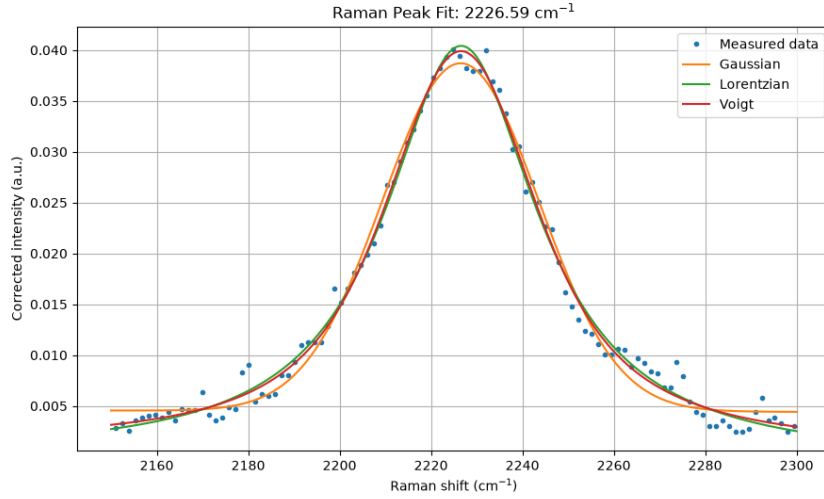


Figure 35: Fit for *peak at 2226 cm⁻¹*.

Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
Gaussian	2226.5 ± 0.2	736.85 ± 0.01	40.2	0.9857
Lorentzian	2226.6 ± 0.2	736.86 ± 0.01	40.9	0.9905
Voigt	2226.6 ± 0.2	736.85 ± 0.01	40.5	0.9912

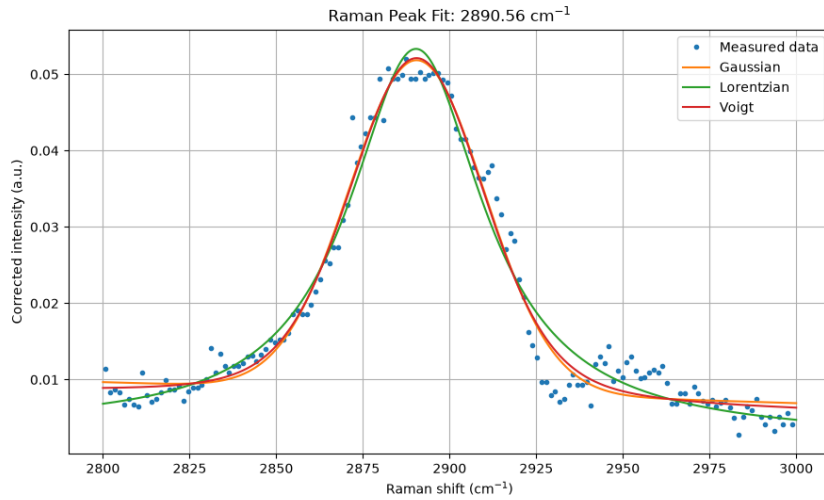


Figure 36: Fit for *peak at 2888 cm⁻¹*.

Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
Gaussian	2890.6 ± 0.3	774.76 ± 0.02	46.0	0.9718
Lorentzian	2890.3 ± 0.4	774.75 ± 0.02	47.2	0.9638
Voigt	2890.6 ± 0.3	774.76 ± 0.02	46.3	0.9727

To find out the content of the unknown sample, the [Search for Species Data by Vibrational Energy Value](#) of the NIST was used.

Table 9: Comparison of the measured Raman bands of the unknown sample with the literature Raman frequencies of liquid acetonitrile (CH_3CN).

Symmetry	Mode	NIST ν 1/cm	Measured ν 1/cm	Deviation 1/cm	Assignment
e	CCN bend	380	450	+70	Tentative
a_1	CC stretch	918	953	+35	Likely
a_1	CH_3 s-deform	1376	1402	+26	Likely
a_1	CN stretch	2249	2226	-23	Likely
a_1	CH_3 s-stretch	2942	2890	-52	Likely

The unknown sample exhibits Raman bands at 450, 953, 1402, 2226, and 2890 cm^{-1} . Comparison with the literature values for liquid acetonitrile shows good correspondence for several of these features. In particular, the bands at 953, 1402, 2226, and 2890 cm^{-1} can be associated with the C–C stretching, CH_3 symmetric deformation, $\text{C}\equiv\text{N}$ stretching, and CH_3 symmetric stretching modes, respectively. The $\text{C}\equiv\text{N}$ stretching band near 2226 cm^{-1} is particularly characteristic of acetonitrile and provides a useful distinction from benzene and CCl_4 . The peak at 450 cm^{-1} lies in the region of the CCN bending mode, although its deviation from the liquid-phase literature value of 380 cm^{-1} is relatively large, making this assignment less certain. Similarly, the measured frequencies show deviations from the literature values, which may result from calibration uncertainties or differences in the experimental conditions. Nevertheless, the overall pattern of several characteristic vibrational modes is consistent with acetonitrile and does not correspond to the characteristic Raman spectra of either benzene or CCl_4 . The unknown sample can therefore be identified as ****consistent with acetonitrile****, while the deviations in individual peak positions should be taken into account when assessing the assignment.

3.4 Ethanol concentration of an unknown sample

The integrated peak area was used as the measure of Raman signal intensity because it accounts for the intensity distributed over the entire spectral band rather than relying only on the peak maximum. This makes the measurement less sensitive to variations in peak width and small shifts in the peak position, providing a more robust quantity for comparing different ethanol concentrations. For this reason, the integrated area is preferable to the peak height for calibration purposes.

The results show a general increase in integrated area with increasing ethanol concentration, particularly for the 100% sample. However, the dependence is not strongly linear, as indicated by the relatively low coefficient of determination of $R^2 = 0.611$. In particular, the integrated areas measured for 25%, 50%, and 75% ethanol are relatively similar, with the 50% sample even giving a slightly lower value than the 25% sample. This suggests that the experimental variations are substantial compared with the concentration-dependent change in Raman intensity. Possible sources include baseline subtraction, measurement noise, variations in the integration range, and fluctuations in the experimental conditions. Therefore, although the integrated area exhibits an overall increase with ethanol concentration, the present measurements do not demonstrate a strong linear calibration relationship.

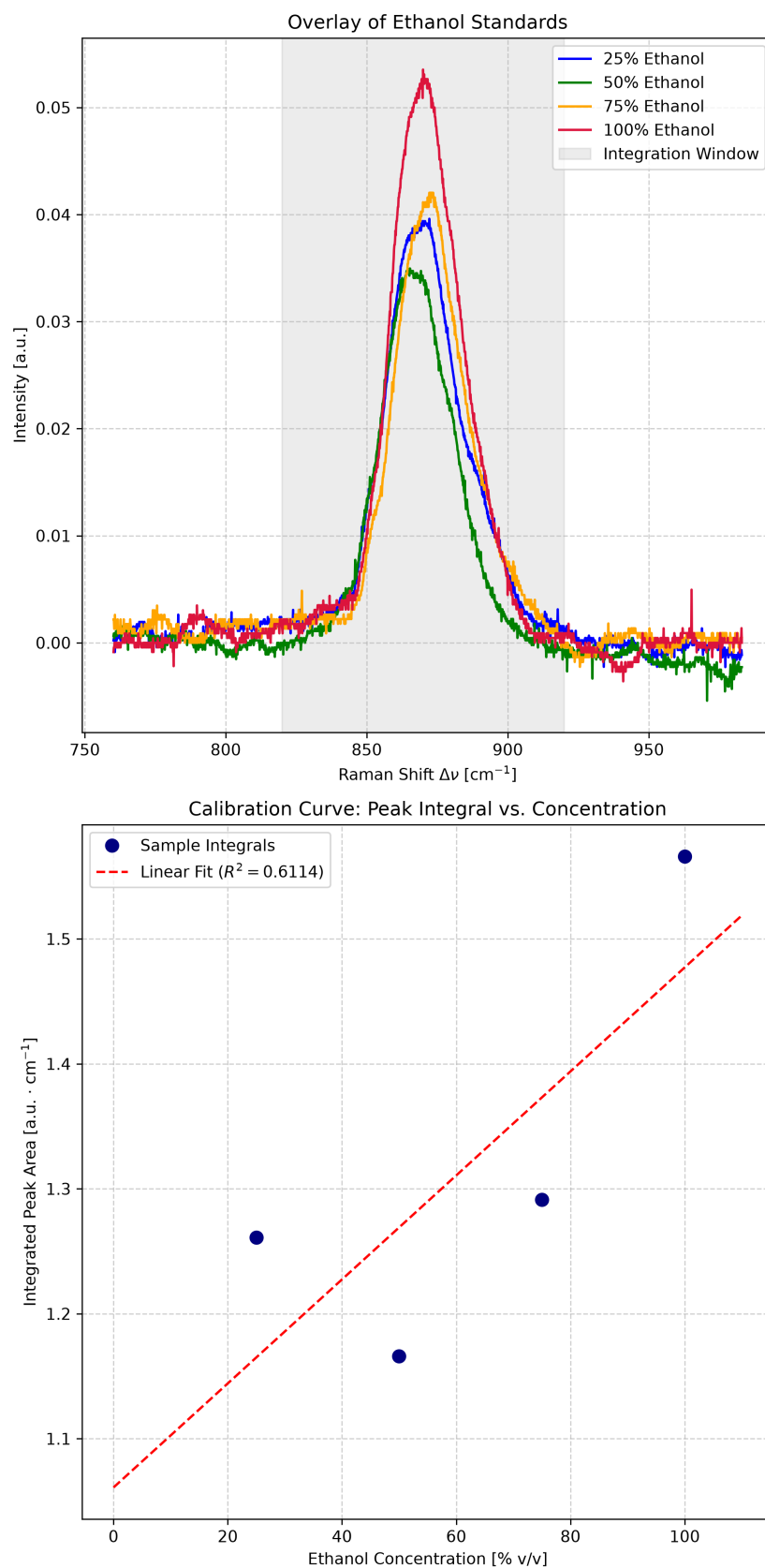


Figure 37: Ethanol spectra for different concentrations (raw)

Table 10: Ethanol Calibration Fit Results

Concentration (%)	Integrated Area
25	1.2609
50	1.1661
75	1.2912
100	1.5661

Linear Fit Statistics
Equation: $\text{Area} = 0.0042 \times C + 1.0610$
 $R^2 = 0.61135$

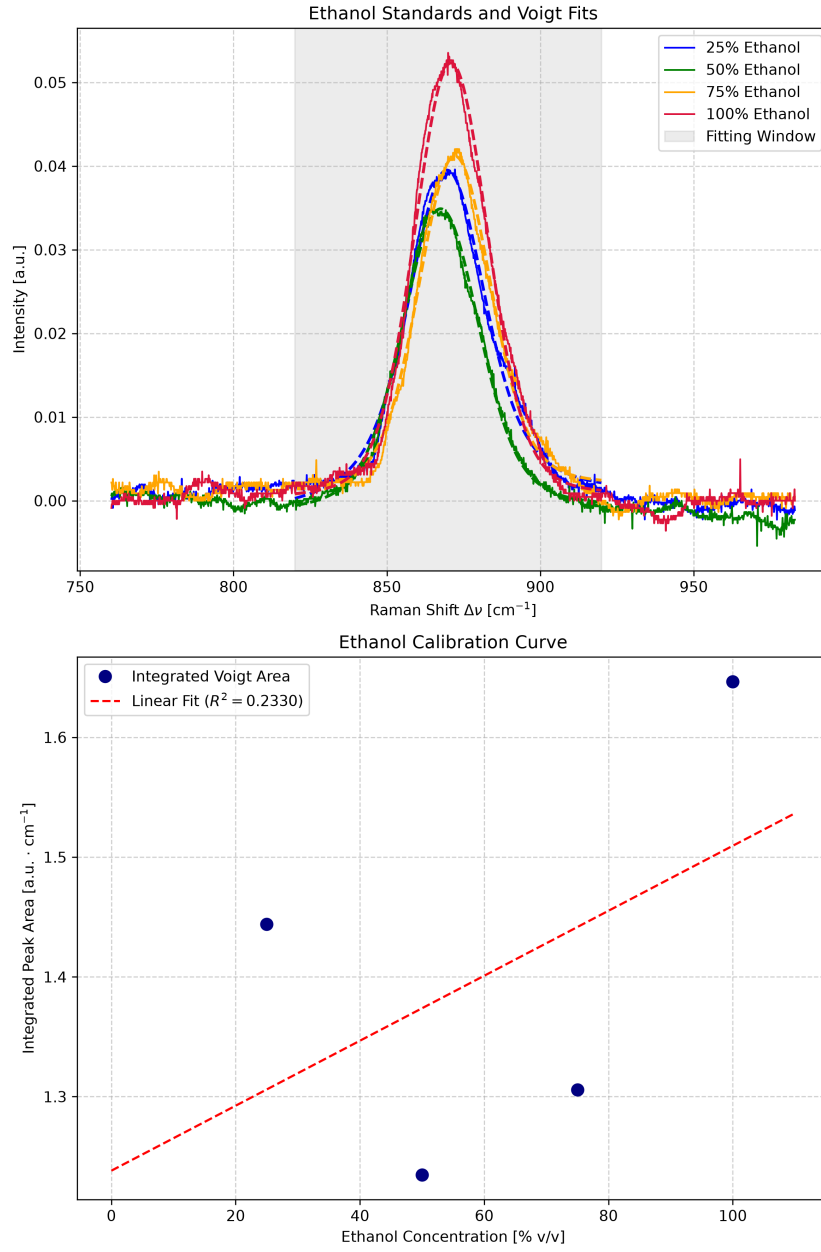


Figure 38: Ethanol spectra for different concentrations (Voigt fit)

Table 11: Ethanol calibration results obtained by integrating the fitted Voigt profiles.

Ethanol concentration (%)	Integrated Voigt area (a.u. cm ⁻¹)
25	1.44
50	1.23
75	1.31
100	1.65
Linear fit	
Equation: $\text{Area} = 0.0027C + 1.2382$	
$R^2 = 0.233$	

It was expected that the integ

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